

Problem 1

- (a) Satisfied. $C=3$ appears in tuples $(4,2,3)$ and $(6,2,3)$, both map to $B=2$. $C=5$ appears only in $(4,2,5)$, mapping to $B=2$. $C=4$ appears only in $(6,2,4)$, mapping to $B=2$. $C=9$ appears only in $(7,8,9)$, mapping to $B=8$. Since every group of tuples sharing the same C value also shares the same B value, $C \rightarrow B$ is satisfied.
- (b) Not satisfied. Counterexample: For $A=4$ and $B=2$, we have tuples $(4,2,3)$ and $(4,2,5)$, the outputs are $C=3$ and $C=5$. Since the C values are different, this is not satisfied.
- (c) Satisfied. $A=4$ appears in tuples $(4,2,3)$ and $(4,2,5)$, both map to $B=2$. $A=6$ appears in tuples $(6,2,3)$ and $(6,2,4)$, both map to $B=2$.
- (d) Not satisfied. Counterexample: For $C=3$, we have tuples $(4,2,3)$ and $(6,2,3)$, the outputs are $A=4$ and $A=6$. Since the A values are different, this is not satisfied.
- (e) Not satisfied. Counterexample: For $(B,C)=(2,3)$, we have tuples $(4,2,3)$ and $(6,2,3)$, the outputs are $A=4$ and $A=6$. Since the A values are different, this is not satisfied.
- (f) Satisfied. This is a trivial functional dependency, because A is already included in the left-hand side. Any dependency of the form: $XYZ \rightarrow X$ is always satisfied, so $ABC \rightarrow A$ is satisfied.
- (g) Not satisfied. $AB \rightarrow AC$ means both $AB \rightarrow A$ (trivially satisfied) and $AB \rightarrow C$ must hold. However, $AB \rightarrow C$ does not hold. Counterexample: tuples $(4,2,3)$ and $(4,2,5)$ both have $(A,B)=(4,2)$, but $C=3$ and $C=5$ respectively. Since the same AB value maps to two different C values, $AB \rightarrow C$ fails, and therefore $AB \rightarrow AC$ is not satisfied.

Problem2

Not equivalent.

① Show every FD in F can be derived from G ($F^+ \subseteq G^+$)

(1) $B \rightarrow CE$:

$B \rightarrow AE$ (given), so $B \rightarrow E$ (decomposition). $E \rightarrow CD$ (given), so $E \rightarrow C$ (decomposition).
Therefore $B \rightarrow C$ (transitivity: $B \rightarrow E, E \rightarrow C$). Combined with $B \rightarrow E$: $B \rightarrow CE$ (union).

(2) $B \rightarrow A$:

$B \rightarrow AE$ (given), so $B \rightarrow A$ (decomposition).

(3) $E \rightarrow D$:

$E \rightarrow CD$ (given), so $E \rightarrow D$ (decomposition).

$\therefore F^+ \subseteq G^+$

② Show that $G^+ \not\subseteq F^+$ (find one FD in G not derivable from F)

(1) $B \rightarrow AE$:

$B \rightarrow CE$ (given), so $B \rightarrow E$ (decomposition). $B \rightarrow A$ (given), so $B \rightarrow AE$ (union).

(2) $E \rightarrow CD$:

$E \rightarrow D$ (given), need $E \rightarrow C$ to get $E \rightarrow CD$. Compute $\{E\}^+$: Start from $\{E\}$, apply $E \rightarrow D$: $\{E, D\}$, no more FDs applicable, $\{E\}^+ = \{E, D\}$, $C \notin \{E\}^+$. There is no dependency deriving $E \rightarrow C$, we can't get $E \rightarrow CD$.

(3) $C \rightarrow B$:

Compute $\{C\}^+$ under F: Start from $\{C\}$, no FD in F has C on the left side. $\{C\}^+ = \{C\}$, $B \notin \{C\}^+$. There is no dependency in F involving C on the left side, so we can't get $C \rightarrow B$.

$\therefore G^+ \not\subseteq F^+$

Since $F^+ \subseteq G^+$ but $G^+ \not\subseteq F^+$, the two sets of functional dependencies are NOT equivalent.

Problem 3

(a) A decomposition $\{R1, R2\}$ is lossless-join iff: $(R1 \cap R2)^+ \supseteq R1-R2$ OR $(R1 \cap R2)^+ \supseteq R2-R1$

Compute the intersection and differences:

	Attributes
$R1 \cap R2$	$\{A\}$
$R1-R2$	$\{B,C\}$
$R2-R1$	$\{D,E\}$

Compute $(R1 \cap R2)^+ = A^+$:

$A \rightarrow BC$ (given)

$B \rightarrow D$ (given)

$CD \rightarrow E$ (given)

$\therefore A^+ = \{A, B, C, D, E\}$

Since $\{A,B,C,D,E\} \supseteq \{B,C\}$ and $\{A,B,C,D,E\} \supseteq \{D,E\}$, $(R1 \cap R2)^+ \supseteq R1-R2$ and $(R1 \cap R2)^+ \supseteq R2-R1$,

$\therefore R1(A, B, C)$ and $R2(A, D, E)$ is a lossless-join decomposition of R (under F).

(b) A decomposition $\{R1, R2\}$ is lossless-join iff: $(R1 \cap R2)^+ \supseteq R1-R2$ OR $(R1 \cap R2)^+ \supseteq R2-R1$

Compute the intersection and differences:

	Attributes
$R1 \cap R2$	$\{C\}$
$R1-R2$	$\{A,B\}$
$R2-R1$	$\{D,E\}$

Compute $(R1 \cap R2)^+ = C^+$:

$A \rightarrow BC$ (given)

$B \rightarrow D$ (given)

$CD \rightarrow E$ (given)

$\therefore C^+ = \{C\}$

$\therefore \{A,B\} \not\supseteq \{C\}$ and $\{D,E\} \not\supseteq \{C\}$, so $R1-R2 \not\supseteq (R1 \cap R2)^+$ and $R2-R1 \not\supseteq (R1 \cap R2)^+$.

$\therefore R1(A, B, C)$ and $R2(C, D, E)$ is not a lossless-join decomposition of R (under F).

(c) BCNF condition: For every non-trivial FD $X \rightarrow Y$ in F^+ , X must be a superkey.

Check each FD for BCNF violation in $R(A,B,C,D,E)$:

FD	Left-side closure	Is left-side a superkey?
$A \rightarrow BC$	$A^+ = \{A,B,C,D,E\}$	Yes
$B \rightarrow D$	$B^+ = \{B,D\}$	No
$CD \rightarrow E$	$CD^+ = \{A,B,C,D,E\}$	Yes
$E \rightarrow A$	$E^+ = \{A,B,C,D,E\}$	Yes

$B \rightarrow D$ violates BCNF (B is not a superkey of R).

Then split R into: (1) $R1(B,D)$ (2) $R2(A,B,C,E)$.

Then check BCNF for (A,B,C,E) . Project F onto $\{A,B,C,E\}$: $A \rightarrow BC$, $E \rightarrow A$. Check keys: $A^+ = \{A,B,C\}$.

But $E \rightarrow A$, so $E^+ = \{A,B,C,E\}$.

Check each FD for BCNF violation in $R2(A,B,C,E)$:

FD	Left-side closure	Is left-side a superkey?
$A \rightarrow BC$	$A^+ = \{A,B,C\}$	No
$E \rightarrow A$	$E^+ = \{A,B,C,E\}$	Yes

So we must decompose using $A \rightarrow BC$.

Then split $R2$ into: (1) $R21(A,B,C)$ (2) $R22(A,E)$.

Check each FD for BCNF violation in R21(A,B,C) and R22(A,E):

FD	Left-side closure	Is left-side a superkey?
$A \rightarrow BC$	$A^+ = \{A,B,C\}$	Yes
$E \rightarrow A$	$E^+ = \{A,E\}$	Yes

For R22(A,E): the only projected FD is $E \rightarrow A$, and $E^+ = \{A,E\} = R22$, so E is a superkey of R22. R22 is in BCNF. Final BCNF decomposition: R1(B,D), R21(A,B,C) and R2(A,E).

Problem 4

(a) A relation is in BCNF if for every non-trivial functional dependency $X \rightarrow Y$, X is a superkey.

Calculate the closures of A,B,C,D,E:

$C^+ = \{C, E, D, B, A\}$, C is a candidate key.

$E^+ = \{E, D, B, C, A\}$, E is a candidate key.

$D^+ = \{D, B, C, A, E\}$, D is a candidate key.

$B^+ = \{B, A, D, C, E\}$, B is a candidate key.

$A^+ = \{A, D, B, C, E\}$, A is a candidate key.

A relation is in BCNF if for every non-trivial FD $X \rightarrow Y$, X is a superkey.

Since the closure of every attribute (A, B, C, D, E) contains all attributes, each attribute is a superkey. Therefore, the relation is in BCNF.

(b)

A decomposition $\{S, T\}$ is lossless-join iff: $(S \cap T)^+ \supseteq S-T$ OR $(S \cap T)^+ \supseteq T-S$

Compute the intersection and differences:

	Attributes
$S \cap T$	$\{D\}$
$S-T$	$\{C, E\}$
$T-S$	$\{A, B\}$

Compute $(S \cap T)^+ = D^+$:

$D \rightarrow BC$ (given)

$B \rightarrow A$ (given)

$A \rightarrow D$ (given)

$C \rightarrow E$ (given)

$E \rightarrow D$ (given)

$\therefore D^+ = \{A, B, C, D, E\}$

$\therefore \{A, B, C, D, E\} \supseteq \{C, E\}$ and $\{A, B, C, D, E\} \supseteq \{A, B\}$.

$\therefore S(C, D, E)$ and $T(A, B, D)$ is a lossless-join decomposition.

(c) Find the non-trivial FDs involving only C, D and E.

From the original FDs:

$C \rightarrow E$

$E \rightarrow D$

$D \rightarrow BC$

Since relation S contains only C, D, and E, we project the dependencies: From $D \rightarrow BC$, we keep:

$D \rightarrow C$

Thus the relevant dependencies are:

$C \rightarrow E$

$E \rightarrow D$

$D \rightarrow C$

These are the non-trivial functional dependencies for $S(C, D, E)$: $C \rightarrow DE$, $D \rightarrow CE$, $E \rightarrow CD$.

(d) Find the non-trivial FDs involving only A, B and D.

From the original FDs:

$B \rightarrow A$

$A \rightarrow D$

$D \rightarrow BC$

Since relation T contains only A, B, and D, we project: From $D \rightarrow BC$, keep: $D \rightarrow B$. Thus the dependencies are: $B \rightarrow A$, $A \rightarrow D$, $D \rightarrow B$. These are the non-trivial functional dependencies for $T(A,B,D)$: $B \rightarrow AD$, $A \rightarrow BD$, $D \rightarrow AB$.

(e)

$C \rightarrow E$: Preserved in $S(C \rightarrow DE)$.

$D \rightarrow BC$: $D \rightarrow B$ preserved in $T(D \rightarrow AB)$, $D \rightarrow C$ preserved in $S(D \rightarrow C)$.

$E \rightarrow D$: Preserved in $S(E \rightarrow CD)$.

$B \rightarrow A$: Preserved in $T(B \rightarrow AD)$.

$A \rightarrow D$: Preserved in $T(A \rightarrow BD)$.

Therefore, the decomposition is dependency preserving.

Problem 5

(a) Since A never appears on RHS, A must be in every candidate key.

①{AB}⁺:

AB → CEF: {A,B,C,E,F}

B → D: {A,B,C,D,E,F} = R

Check subsets: {A}⁺ = {A}, {B}⁺ = {B,D}

Neither subset is a superkey → AB is a candidate key

②{AC}⁺:

AC → F: {A,C,F}

No more FDs applicable

{AC}⁺ = {A,C,F} ≠ R

③{AE}⁺:

No FDs applicable

{AE}⁺ = {A,E} ≠ R

④{AF}⁺:

No FDs applicable

{AF}⁺ = {A,F} ≠ R

Since pairs {AC}, {AE}, {AF} are not superkeys, then try triples:

①{ACE}⁺:

ACE → B: {A,B,C,E}

AB → CEF: {A,B,C,E,F}

B → D: {A,B,C,D,E,F} = R

Check subsets: {AC}⁺={A,C,F}, {AE}⁺={A,E}, {CE}⁺={C,E}

No proper subset is a superkey → ACE is a candidate key

②{AEF}⁺:

AEF → BC: {A,B,C,E,F}

B → D: {A,B,C,D,E,F} = R

Check subsets: {AE}⁺={A,E}, {AF}⁺={A,F}, {EF}⁺={E,F}

No proper subset is a superkey → AEF is a candidate key

③{ACF}⁺:

AC → F: already have F

{ACF}⁺ = {A,C,F} ≠ R

Therefore, the candidate keys for R(A,B,C,D,E,F) are: {AB}, {ACE}, {AEF}.

(b) A relation is in 3NF if for every non-trivial FD $X \rightarrow Y$: (1) X is a superkey (2) Y is a prime attribute(part of a candidate key). Prime attributes: A, B, C, E, F.

Check each FD:

AC → F: F is prime, then it is in 3NF.

B → D: D is non-prime, B is not a superkey, (B⁺ = {B,D}), then it is not in 3NF.

AB → CEF: AB is superkey, then it is in 3NF.

ACE → B: ACE is superkey, then it is in 3NF.

AEF → BC: AEF is superkey, then it is in 3NF.

Therefore, R is not in 3NF.

(c) A relation is in BCNF if for every non-trivial functional dependency $X \rightarrow Y$, X is a superkey.

AC → F: AC is not superkey, then it is not in BCNF.

$B \rightarrow D$: B is not a superkey, then it is not in BCNF.

$AB \rightarrow CEF$: AB is superkey, then it is in BCNF.

$ACE \rightarrow B$: ACE is superkey, then it is in BCNF.

$AEF \rightarrow BC$: AEF is superkey, then it is in BCNF.

Therefore, R is not in BCNF.

(d) BCNF condition: For every non-trivial FD $X \rightarrow Y$, X must be a superkey.

Check each FD for BCNF violation in $R(A,B,C,D,E,F)$:

FD	Left-side closure	Is left-side a superkey?
$AC \rightarrow F$	$AC^+ = \{A,C,F\}$	No
$B \rightarrow D$	$B^+ = \{B,D\}$	No
$AB \rightarrow CEF$	$AB^+ = \{A,B,C,D,E,F\}$	Yes
$ACE \rightarrow B$	$ACE^+ = \{A,B,C,D,E,F\}$	Yes
$AEF \rightarrow BC$	$AEF^+ = \{A,B,C,D,E,F\}$	Yes

$B \rightarrow D$ violates BCNF (B is not a superkey of R).

Then split R into: (1) $R_1(B,D)$ (2) $R_2(A,B,C,E,F)$. Since $B \rightarrow D$, B is superkey of R_1 , so R_1 is in BCNF.

Then check each FD for BCNF violation in $R(A,B,C,E,F)$:

FD	Left-side closure	Is left-side a superkey?
$AC \rightarrow F$	$AC^+ = \{A,C,F\}$	No
$AB \rightarrow CEF$	$AB^+ = \{A,B,C,E,F\}$	Yes
$ACE \rightarrow B$	$ACE^+ = \{A,B,C,E,F\}$	Yes
$AEF \rightarrow BC$	$AEF^+ = \{A,B,C,E,F\}$	Yes

Then split R_2 into: (1) $R_{21}(A,C,F)$ (2) $R_{22}(A,B,C,E)$. Since $AC \rightarrow F$, AC is superkey of R_{21} , so R_{21} is in BCNF. Since $AB \rightarrow CE$, $E \rightarrow A$, compute closures: $AB^+ = \{A,B,C,E\} \Rightarrow AB$ is a superkey, $E^+ = \{E,A,B,C\} \Rightarrow E$ is a superkey. Since all non-trivial FDs have superkey on the left-hand side, R_{22} is in BCNF.

Final BCNF decomposition: $R_1(B,D)$, $R_{21}(A,C,F)$ and $R_{22}(A,B,C,E)$.